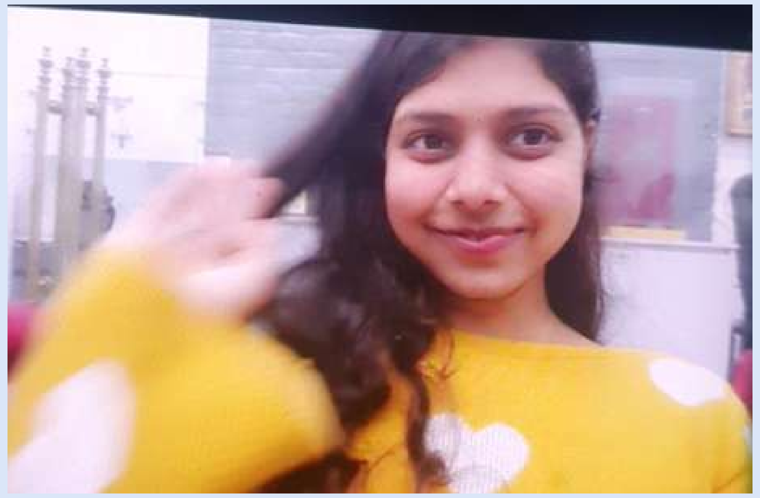
TEAM NAME :- NodeVS

Healthsphere

AI-Powered Smart Healthcare Companion

MEET THE TEAM



ASHMITA GOYAL

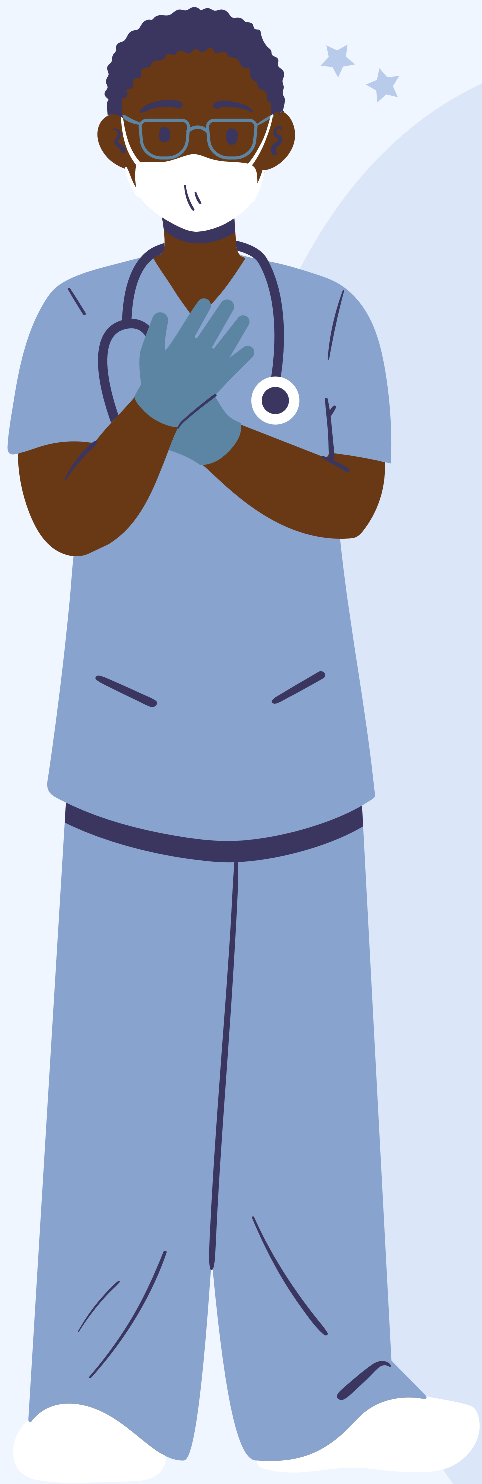
Role: Full Stack Developer & AI Integration

NEERAJ MISHRA

Role: Backend Developer & System Design

Problem Statement

Healthcare systems today are fragmented, slow, and inaccessible, leading to delayed medical decisions, poor emergency response, and lack of reliable health guidance especially in rural and underserved areas.



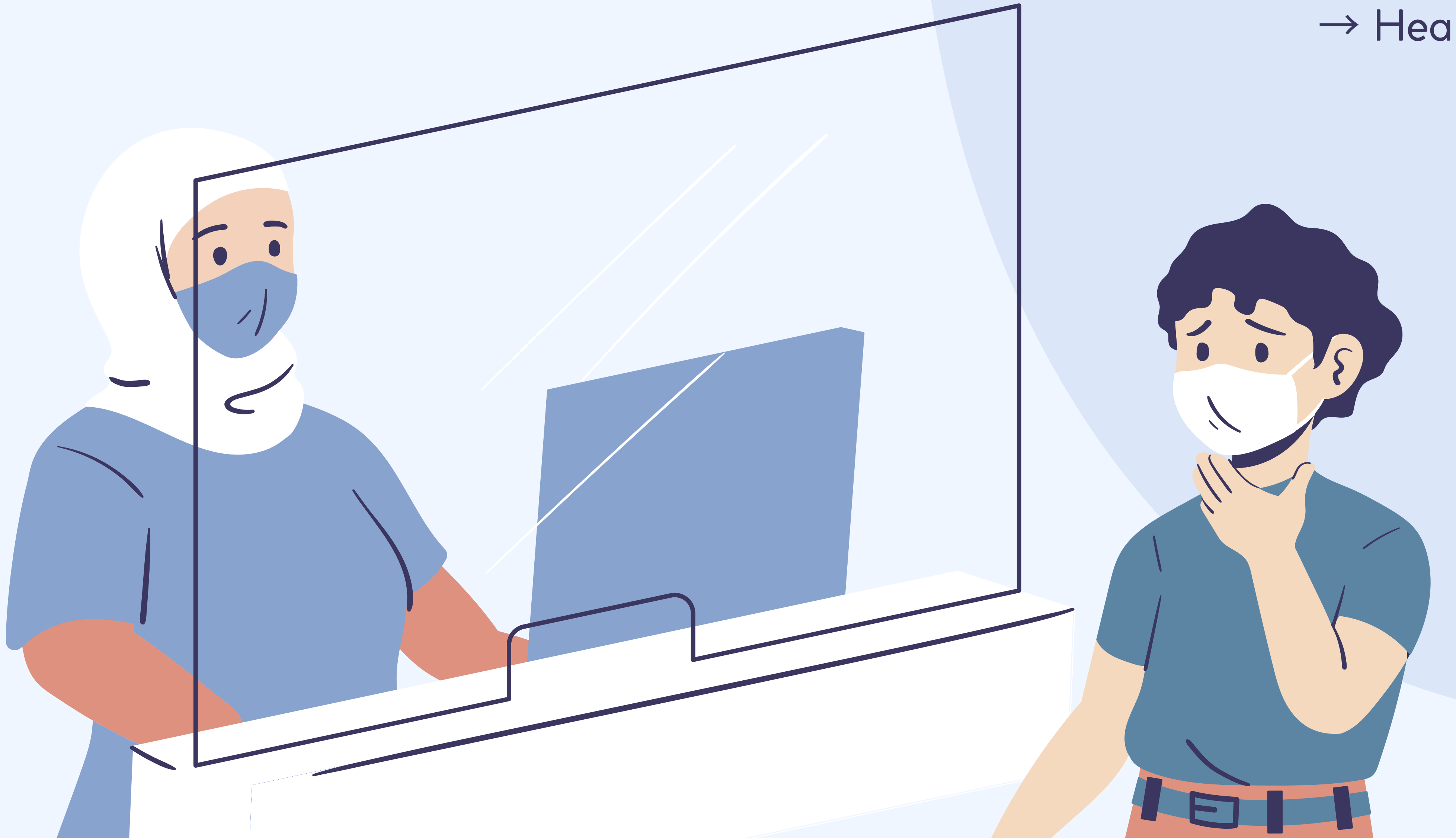
Problem Statement Expansion:

- Huge gap between urban and rural healthcare access
- Limited reach of telemedicine and expert doctors
- Overdependence on unverified online medical information
- Lack of real-time emergency response systems
- No centralized platform for medical records & health tracking
- Shortage of healthcare professionals → overburdened system
- Absence of AI-driven preventive healthcare & early detection
- No systems for:
- Disease outbreak prediction
- Fraudulent health claim detection
- Caregiver support

How healthcare problems appear and affect real users:-

How it manifests for the user or client

· Confusion due to unverified medical advice
· Delayed diagnosis & lack of instant guidance
· No immediate help during emergencies
· Scattered or missing medical records
· Missed medicines & routine checkups
· Limited access in rural areas
→ Healthcare feels slow, disconnected, and unreliable



How it impacts the user or client

Increased risk of serious health complications
· Higher treatment costs due to delays
· Overburdened hospitals & systems
· Poor health management & lifestyle
· Mental stress & uncertainty
· Higher chances of wrong decisions
→ Results in reactive, inefficient healthcare system

USER PERSONA & DESIGN MINDSET

A user persona represents real users by capturing their needs, behaviors, challenges, and goals. It helps in designing solutions that are practical, relatable, and aligned with real-world usage rather than assumptions.

Design Mindset

- Build for **real users**, not hypothetical scenarios
- Focus on **daily life challenges and pain points**
- Prioritize **need and necessity over convenience**
- Design for **constraints like limited access, time, and resources**
- Ensure **simplicity, empathy, and ease of use**
- Create solutions that **positively impact everyday life**

Approach

By understanding users deeply, we design solutions that are **accessible, reliable, user-friendly, and capable of making meaningful improvements in real-world healthcare experiences.**

USER PERSONA 1

Ramesh Kumar

Age: 35 | Location Rural Area

Occupation Farmer

ABOUT THE USER

Simple, hardworking; and family–onemed individual with limitted access to nearby heatlticarefacilities and dioltal fools.

PROBLEMS

- No nearby hospitals or doctors
- Relles on unverrtied local remedies
- Faces delays during meatcal emergancies
- Language barrier with digitill plafforms

CHALLENGES

- Difficutly accessing ftmely medical help
- Limited intemel connectivity
- Financial constrants for healthcare

GOALS & NEEDS

- Ouick and rellable health guidance
- Fmergency support with location tracking
- Simple, local language interfase
- Affordable and accessible healthcare solutions

USER PERSONA 2

Priya Sharma

Age: 28 | Location: Urban City | Occupation: Working Professional

ABOUT THE USER

Ambitious, career-focused individual managing a busy lifestyle with limited time for regular health checkups and self-care.

CHALLENGES

- Balancing work and personal health
- Lack of consistent health monitoring
- Ignoring early symptoms due to busy schedule
- Stress and lifestyle-related health risks

PROBLEMS

- Frequently misses medications & checkups
- Relies on quick online searches for symptoms
- No time for doctor visits
- Struggles with maintaining a healthy routine

GOALS & NEEDS

- Quick and reliable **AI health assistance**
- Smart medicine & routine reminders
- Easy access to nearby healthcare services
- Efficient and time-saving healthcare solutions

USER PERSONA 3

Shanti Devi

Age: 65 | Location: Semi-Urban Area | Occupation: Retired

ABOUT THE USER

Elderly individual with ongoing health issues who depends on family members for medical support and struggles with using modern digital tools.

CHALLENGES

- Forgetting medicines & checkups
- Difficulty using complex interfaces
- Lack of continuous health monitoring
- Mobility issues affecting hospital visits

PROBLEMS

- Difficulty managing multiple medications
- No real-time help during health emergencies
- Limited understanding of digital healthcare apps
- Relies heavily on family for support

GOALS & NEEDS

- Simple and easy-to-use interface
- Voice-based assistance for interaction
- Timely medicine reminders & alerts
- Caregiver support & emergency assistance

HEALTHSPHERE

Description

HealthSphere AI is an AI-powered healthcare platform that provides real-time medical guidance, emergency support, and health monitoring in one unified system.

Core Idea

One intelligent platform for healthcare guidance, emergency response, medical records, and preventive care — accessible anytime, anywhere

Key Features

AI assistant, SOS emergency system, medicine reminders, health tracking dashboard, and secure medical records vault.

Demographic Impact

Designed for general users, rural populations, elderly patients, caregivers, and individuals needing accessible and reliable healthcare support.

SOLUTION

Solution Overview

HealthSphere AI — an intelligent healthcare platform integrating AI guidance, emergency response, health tracking, and secure medical records into one unified system.

Target Users

General users seeking daily health support
Rural populations with limited access to doctors
Elderly patients and caregivers
Individuals with ongoing health conditions

Problem Addressed

- Fragmented healthcare services
- Delayed emergency response
- Lack of trusted medical guidance
- Limited accessibility in underserved areas

SOLUTION

How It Works

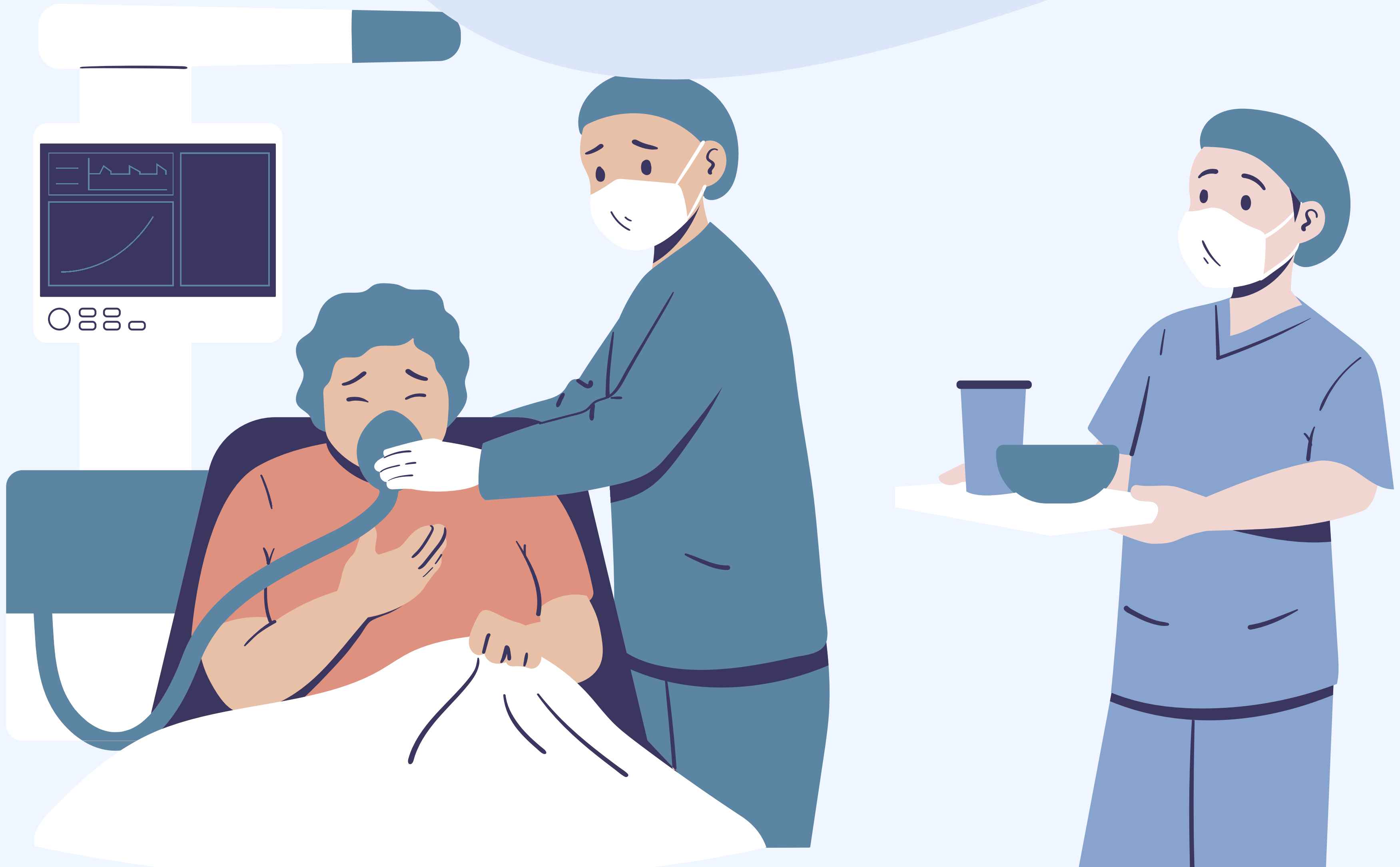
Users interact with an AI assistant via chat or voice to receive instant medical guidance, track health data, manage medications, and access emergency services with real-time location support.

Value Proposition

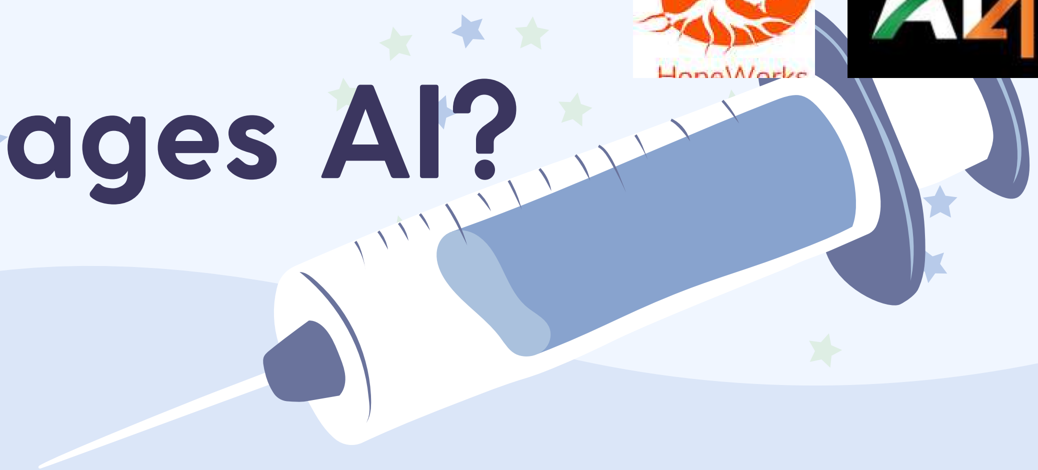
It enables a shift from reactive to proactive healthcare, making reliable, accessible, and real-time medical support available to everyone

Impact

- Faster and more informed medical decisions
- Improved emergency response time
- Better health monitoring and prevention
- Reduced burden on healthcare systems



How your proposed solution Leverages AI?



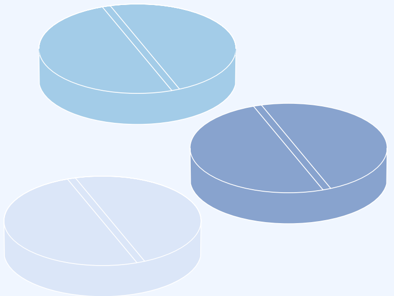
AI Technologies Used

- **GPT-based Assistant** → Real-time medical guidance
- **RAG Framework** → Verified, context-aware responses
- **Whisper API** → Voice-based interaction
- **OCR (Vision AI)** → Medical report digitization
- **Predictive Models** → Health risk analysis

Key Benefits

- Faster and informed decision-making
- Improved accuracy and reliability
- Reduced dependency on unverified sources
- Enhanced accessibility through voice support

AI assists healthcare professionals, not replaces them



PROJECT DETAILS

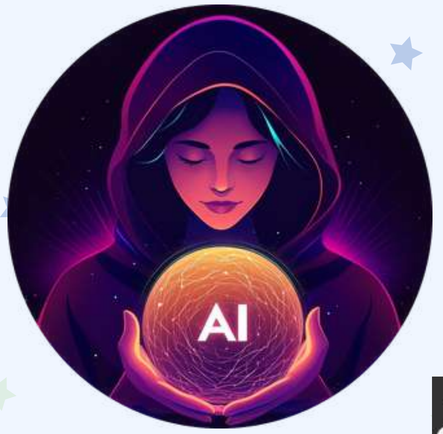
HealthSphere AI is a unified healthcare ecosystem that integrates AI intelligence, real-time emergency systems, and health data management into a scalable and secure platform.

Core Functional Modules

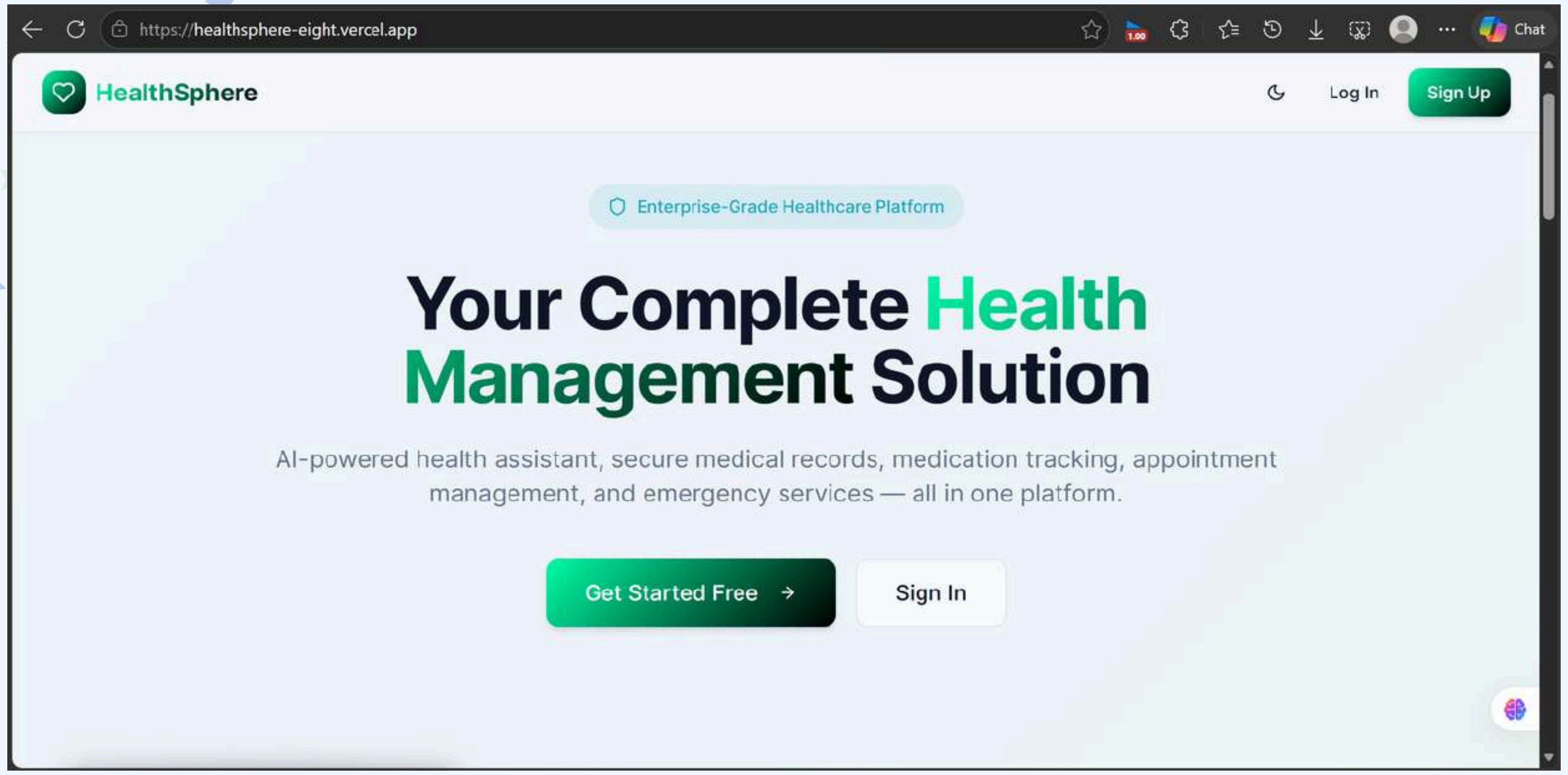
- AI Health Engine → Symptom analysis, risk scoring, first-aid guidance
- Emergency Response System → SOS alerts, live location tracking, rapid assistance
- Health Management System → Daily logs, routine tracking, predictive insights
- Medical Records System → Secure storage, OCR-based extraction, categorization
- Service Discovery Module → Nearby hospitals, pharmacies, blood banks

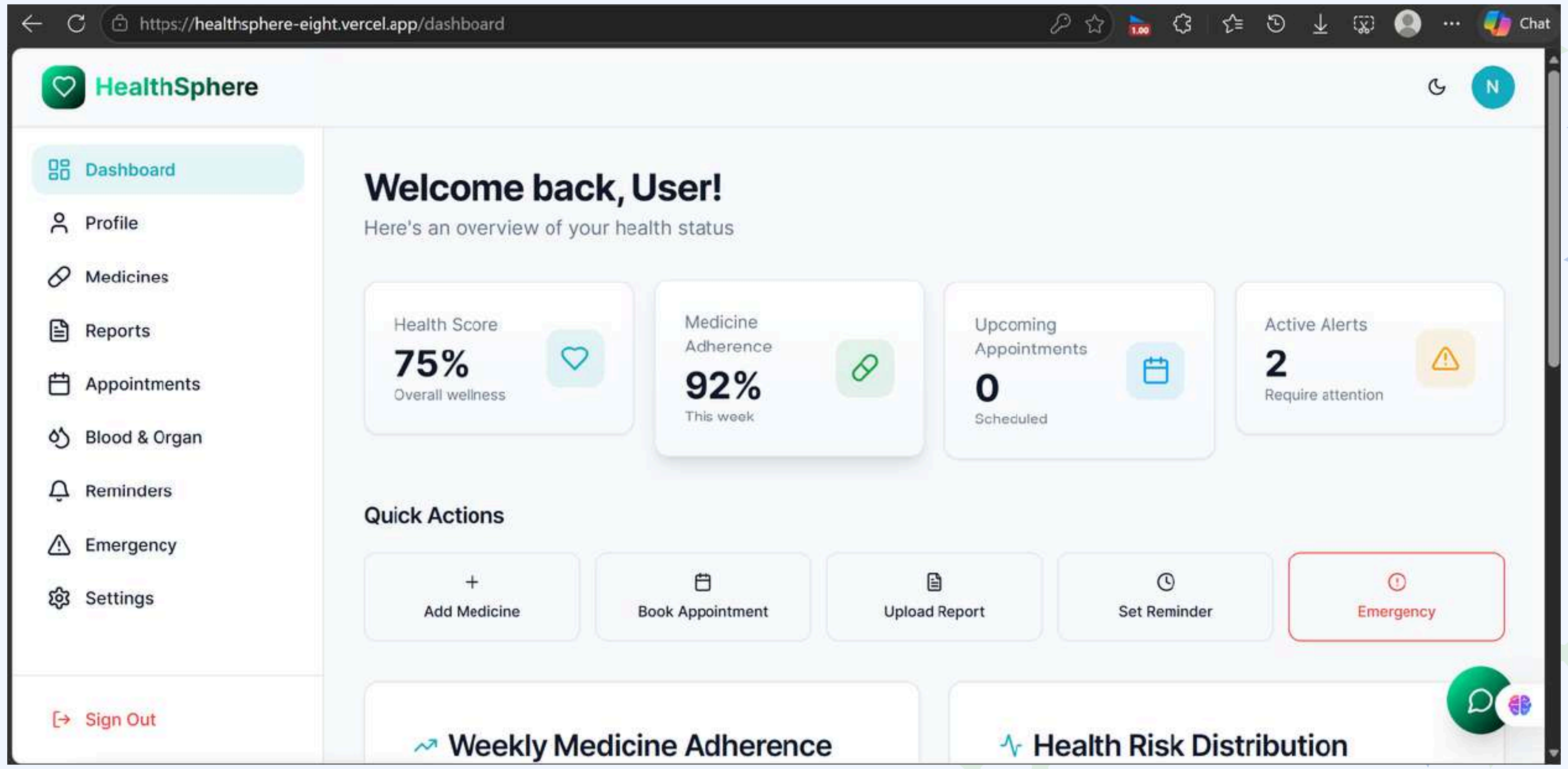
Key Highlights

- Real-time decision support
- Multilingual & voice-enabled interaction
- Designed for both urban & rural accessibility
- Scalable for large user base



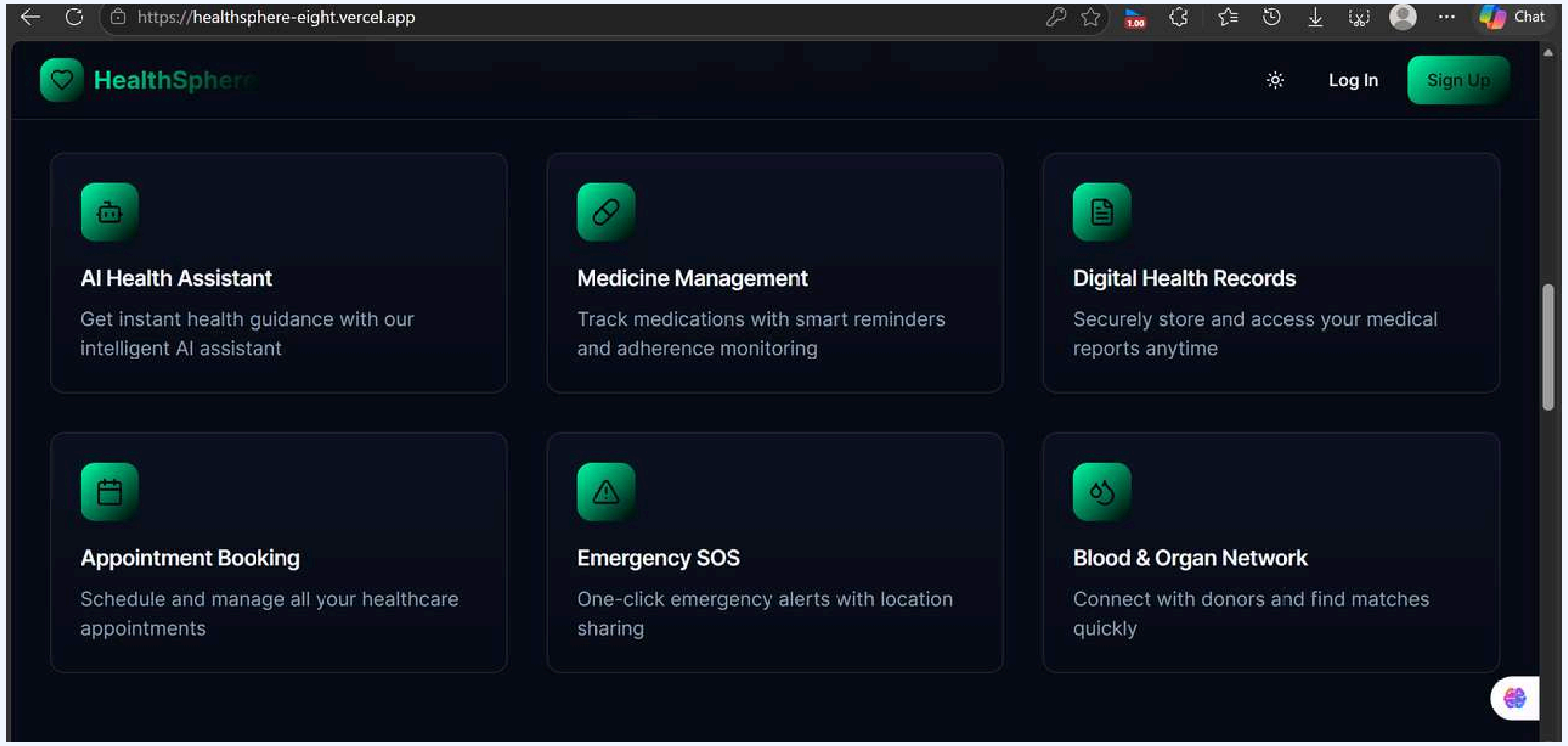
WORKING PROTOTYPE



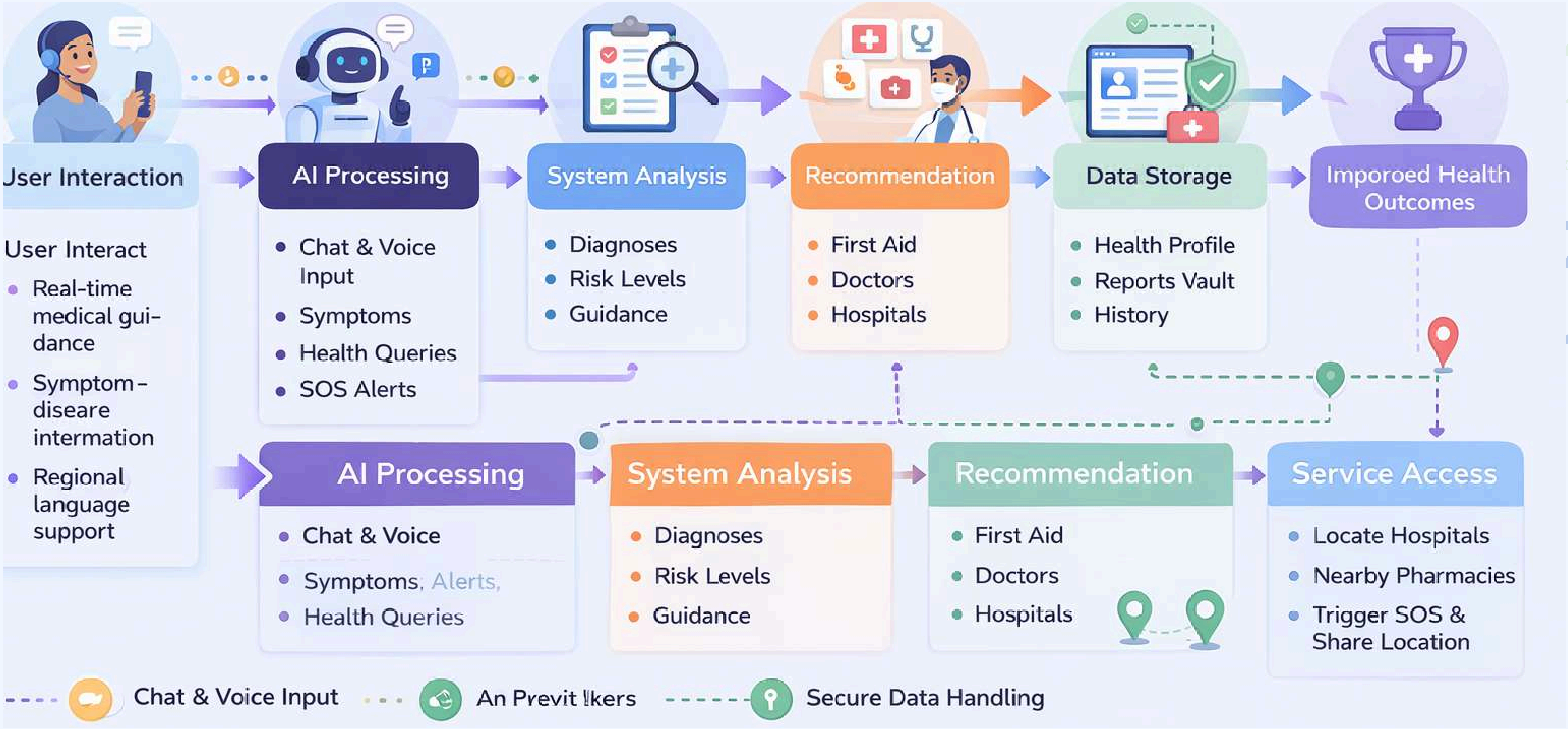


[HealthSphere](#)

[Github link](#)



User Flow (How System Works)



System Architecture

System Architecture Overview

Frontend Layer
- Next.is
- Tallwind CSS

Backend Layer
- Nodisis
- Express
- JWT Auth

Backend Layer
- Node js
- Express
- JWT Auth

AI Layer
- GPT. Whbper AP)
- Vision AI (OCR)
- Prediction Modsss

Database Layer
- PostgreGOL
- User Trata & Health Recorcis

Integration Layer
- Google Maps AP)
- Emergency Ain
- Cloud Storage

Cloud Infrastructure

Chat & Voice Input · Anonyrnized Inputs · Secure Data Handling

Technical Request for Facilitated Datasets

- **Specification of Required Data**

The system requires anonymized patient health records and symptom-based datasets to enable accurate AI-driven diagnosis and recommendations. It also needs structured symptom–disease mapping data, regional language datasets for multilingual support, and real-time hospital and emergency service location data. Additionally, medical reports in the form of PDFs and images are required for OCR-based extraction and intelligent categorization within the platform.

- **The "Ask" vs. "Leveraged" Mapping**

Currently, the platform leverages OpenAI APIs for AI-based medical assistance, publicly available healthcare datasets for initial model training, and Google Maps APIs for location-based services. However, to improve accuracy and real-world applicability, the system requires access to verified clinical datasets, government healthcare APIs for hospitals and schemes, and region-specific health and demographic data to ensure localized and inclusive healthcare delivery.

Technical Justification

The inclusion of high-quality and verified datasets significantly enhances the accuracy and reliability of AI-generated responses. It enables real-time healthcare insights, improves predictive analytics capabilities, and ensures that the system performs effectively across diverse populations, especially in rural and underserved regions. These datasets are essential for building a trustworthy, scalable, and clinically relevant healthcare solution.

Environment & Security Requirements

The platform is designed to operate within a secure cloud-based infrastructure that ensures scalability and high availability. It implements end-to-end encryption to protect sensitive health data, along with robust authentication and access control mechanisms. A consent-based data sharing model is followed to maintain user privacy, and the system is built to align with healthcare data protection standards such as HIPAA and GDPR, ensuring compliance and trustworthiness.

ANTICIPATED RESULTS

Persona Benefit

Rural users and elderly patients benefit the most, as the platform provides instant medical guidance, multilingual support, and real-time emergency assistance in areas with limited healthcare access. This helps in reducing diagnosis delays by approximately 40–50%, improving early intervention and overall health outcomes.

Work Impact on Stakeholders

Healthcare providers may experience a slight increase in digital interactions (10–15%) due to AI-assisted consultations and data inputs. However, this is balanced by reduced manual workload, better patient prioritization, and improved efficiency in managing cases and medical records.

Solution Scalability

The system is built on a scalable cloud-based and modular architecture, allowing seamless expansion across regions and user groups. It supports integration with hospitals, government health systems, and wearable devices, while efficiently handling large-scale data and user traffic in real-world deployment.

Conclusion

HealthSphere AI transforms healthcare from reactive to proactive by combining AI intelligence, real-time emergency response, and health management into one unified platform.

It enables faster medical decisions, improves accessibility, and promotes preventive care while reducing reliance on unreliable medical sources.

With its scalable and user-centric design, HealthSphere AI has the potential to bridge healthcare gaps and deliver reliable, real-time support to people anytime, anywhere.

Making intelligent, accessible, and real-time healthcare available for everyone

Thank you for your attention